# Individual Reflection on Group Project

**Project Title:** Edge Deployable Rock Detection and Burial Status Classification



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## 1. Your views on the completing the project as a team

The project brought together the dataset quality, model development and edge deployment together, which was useful for working as a team. My job at the centre was image annotation and quality checking, as well as converting and transferring files and deploying YOLOv8n as ONNX FP32, TFLite INT8 and NCNN FP16. These tasks depended on outputs from other members and were used directly into the final comparison. I learned that in order to have reliable teamwork, you need traceable files, consistent evaluation settings and early communication of runtime issues, because an annotation error or any inconsistency could affect the overall conclusion.

## 2. Your views on project management and execution of your team

Our work went in a well defined manner from annotation audit to model selection, conversion and Raspberry Pi testing. The audit resulted in 448 valid image lable pairs and 706 rocks labelled with no outstanding annotation errors. Once YOLOv8n was chosen, all the representations were evaluated with the same 90 images, $640 \times 640$ workload with smaller smoke-test subset for functional checks. For my assigned task, I made conversions and transferred models to the Raspberry Pi, verified inference, addressed runtime problems and documented results. This taught me that export success is only the first stage and deployment needs to be verified on the target device under a controlled workload.

## 3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** Annotation checking resulted in a consistent dataset for further evaluation and the fixed Raspberry Pi workload made the runtime comparison fair. TFLite INT8 had a mean latency of 44.13ms and 22.62 FPS, NCNN FP16 reached 13.82 FPS with stronger mAP50 than INT8. This showed the accuracy and efficiency trade off between representations.
- **What Did Not Work Well:** ONNX FP32 was able to make an inference but failed to complete the valid benchmark and more attempts resulted in reboots. I didn't find out if it was a thermal, power, memory, or runtime issue, but I learned to never assign any unsupported explanation. Continuous stability evidence was also uneven across formats.
- **What I Would Do Differently:** I would define the complete deployment test matrix earlier and run continuous tests of equal length, for each assigned format. I'd take notes on CPU frequency, throttle flags, power, etc, and include a controlled active-cooling comparison, and add a second-person annotation review for difficult half-buried examples

## 4. Any other reflections

In this project, I learned more about the practical aspects of deployment to the edge. I learnt how to deal with model export, smoke test compatibility, completion of the 90 image benchmark and sustained operation. Reporting negative results, such as the incomplete ONNX benchmark, was also important for an honest comparison. This experience improved my skills in annotation quality control, model conversion, Raspberry Pi troubleshooting and evidence-based evaluation.